

Fitting Covariance vs Response

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We fit connectivity in two ways: by fitting the raw responses, and fitting the covariance of the responses. How are the resulting connectivities related?

When we fit the responses, our loss is

$$\mathcal{L}_{\text{resp}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y} - \mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2,$$

while when fitting covariances, we minimize

$$\mathcal{L}_{\text{cov}}(\mathbf{Z}) = \frac{1}{2} \|\mathbf{Y}^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T \mathbf{Z}^T \mathbf{J} \mathbf{Z} \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{Z} - \mathbf{I}\|_F^2.$$

To relate these, we use the decomposition into mean-subtracted and mean components.

$$\mathbf{X} = \mathbf{J}\mathbf{X} + (\mathbf{I} - \mathbf{J})\mathbf{X} \triangleq \tilde{\mathbf{X}} + \bar{\mathbf{X}}.$$

Notice that these two components are orthogonal,

$$\langle \tilde{\mathbf{X}}, \bar{\mathbf{X}} \rangle = \text{tr}(\tilde{\mathbf{X}}^T \bar{\mathbf{X}}) = \text{tr}(\mathbf{X}^T \mathbf{J}^T (\mathbf{I} - \mathbf{J}) \mathbf{X}) = \text{tr}(\mathbf{X}^T \mathbf{0} \mathbf{X}) = 0.$$

That means that we can decompose our response loss into two independent components,

$$\begin{aligned} \mathcal{L}_{\text{resp}}(\mathbf{Z}) = \mathcal{L}_{\text{resp}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|\mathbf{J}\mathbf{Y} - \mathbf{J}\mathbf{Z}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{1}{2} \|\bar{\mathbf{Y}} - \bar{\mathbf{Z}}\mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

We can do the same thing for the covariance loss,

$$\begin{aligned} \mathcal{L}_{\text{cov}}(\mathbf{Z}) = \mathcal{L}_{\text{cov}}(\mathbf{J}\mathbf{Z}, \bar{\mathbf{Z}}) &= \frac{1}{2} \|(\mathbf{J}\mathbf{Y})^T \mathbf{J} \mathbf{Y} - \mathbf{X}^T (\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) \mathbf{X}\|_F^2 + \frac{\lambda}{2} \|\mathbf{J}\mathbf{Z} - \mathbf{J}\|_F^2 \\ &\quad + \frac{\lambda}{2} \|\bar{\mathbf{Z}} - \bar{\mathbf{I}}\|_F^2. \end{aligned}$$

From this, we immediately see that the regularization in the covariance case will send $\bar{\mathbf{Z}}$ to $\bar{\mathbf{I}}$.

Therefore, the aspect of the connectivity to compare between the two losses is $\mathbf{J}\mathbf{Z}$. We can polar decompose this into the product of a pure rotation $\mathbf{R}$ and a positive-semi-definite stretch $\mathbf{S}$,

$$\mathbf{J}\mathbf{Z} = \mathbf{R}\mathbf{S}.$$

Since $(\mathbf{J}\mathbf{Z})^T (\mathbf{J}\mathbf{Z}) = \mathbf{S}^T \mathbf{R}^T \mathbf{R} \mathbf{S} = \mathbf{S}^T \mathbf{S}$, we can see that the covariance fit only determines the stretch component, while the response fit determines both.

Therefore, when comparing connectivity solutions, we should use $\mathbf{R}$ and $\mathbf{S}$ computed from $\mathbf{J}\mathbf{Z}$, not $\mathbf{Z}$, when determine the effects of each component.

Response Fits

Returning to the response fits, we find that Free fit, and its symmetrically constrained version, give the same performance. In fact, the symmetrically constrained one might be a little better. What do the fits look like?

The free fits is easy. The gradient is

$$\nabla \mathcal{L}_{\text{resp}} = -(\mathbf{Y} - \mathbf{Z}\mathbf{X})\mathbf{X}^T + \lambda(\mathbf{Z} - \mathbf{I}).$$

Setting this to zero gives

$$\mathbf{Z}_{\text{free}} = (\mathbf{X}\mathbf{X}^T + \lambda\mathbf{I})^{-1}(\mathbf{Y}\mathbf{X}^T + \lambda\mathbf{I}).$$

When we constrain them to be symmetric, we add a Lagrange multiplier to the loss, to get

$$\mathcal{L}_{\text{sym}} = \mathcal{L}_{\text{resp}} + \text{tr}(\mathbf{\Lambda}^T(\mathbf{Z} - \mathbf{Z}^T)).$$

The gradient then becomes

$$\begin{aligned}\nabla \mathcal{L}_{\text{sym}} &= -(\mathbf{Y} - \mathbf{Z}\mathbf{X})\mathbf{X}^T + \lambda(\mathbf{Z} - \mathbf{I}) + \mathbf{\Lambda} - \mathbf{\Lambda}^T. \\ \nabla \mathcal{L}_{\text{sym}}^T &= -\mathbf{X}(\mathbf{Y}^T - \mathbf{X}^T\mathbf{Z}) + \lambda(\mathbf{Z} - \mathbf{I}) + \mathbf{\Lambda}^T - \mathbf{\Lambda}.\end{aligned}$$

Adding them when the gradient is zero eliminates the Lagrange multipliers and gives

$$\mathbf{X}\mathbf{Y}^T + \mathbf{Y}\mathbf{X}^T + 2\lambda\mathbf{I} = \mathbf{Z}\mathbf{X}\mathbf{X}^T + \mathbf{X}\mathbf{X}^T\mathbf{Z} + 2\lambda\mathbf{Z}.$$

If we define

$$\mathbf{\Sigma}_{xx} \triangleq \mathbf{X}\mathbf{X}^T + \lambda\mathbf{I}, \quad \mathbf{\Sigma}_{xy} \triangleq \mathbf{X}\mathbf{Y}^T + \lambda\mathbf{I}, \quad \mathbf{\Sigma}_{yx} \triangleq \mathbf{Y}\mathbf{X}^T + \lambda\mathbf{I},$$

we get

$$\mathbf{\Sigma}_{xy} + \mathbf{\Sigma}_{yx} = \mathbf{Z}\mathbf{\Sigma}_{xx} + \mathbf{\Sigma}_{xx}\mathbf{Z}.$$

$\mathbf{\Sigma}_{xx}$ is invertible, so we get equivalently

$$\mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{xy} + \mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{yx} = \mathbf{\Sigma}_{xx}^{-1}\mathbf{Z}\mathbf{\Sigma}_{xx} + \mathbf{Z}.$$

Recognizing the second term on the left-hand side as the free solution, gives a relation between the two as

$$\mathbf{\Sigma}_{xx}^{-1}\mathbf{\Sigma}_{xy} + \mathbf{Z}_{\text{free}} = \mathbf{\Sigma}_{xx}^{-1}\mathbf{Z}_{\text{sym}}\mathbf{\Sigma}_{xx} + \mathbf{Z}_{\text{sym}}.$$

Rearranging, we can express $\mathbf{Z}_{\text{free}}$ in terms of $\mathbf{Z}_{\text{sym}}$ as

$$\mathbf{Z}_{\text{free}} = \mathbf{\Sigma}_{xx}^{-1}(\mathbf{Z}_{\text{sym}}\mathbf{\Sigma}_{xx} - \mathbf{\Sigma}_{xy}) + \mathbf{Z}_{\text{sym}}.$$

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